Group Cohomology

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Contents

1 **Definitions and resolutions** 1 Lecture 1

1 Definitions and resolutions

Let G be a group.

Definition 1.1 (integral group ring)
The *integral group ring* $\mathbb{Z}G$ has elements being formal sums

$$\sum n_g g$$

for $n_g \in \mathbb{Z}, g \in G$, and only finitely many n_g are non-zero. This is a free abelian group under addition, i.e.

$$\left(\sum m_g g\right) + \left(\sum n_g g\right) = \sum (m_g + n_g) g$$

and we have multiplication

$$\left(\sum m_h h\right)\left(\sum n_k k\right) = \sum_g \left(\sum_{hk=g} m_h n_k\right) g$$

This is the associative ring underlying the integral representation theory of G. We will write $1 = 1e$ for the multiplicative identity in G.

Definition 1.2 (module)
A *(left) $\mathbb{Z}G$-module* M is an abelian group under addition, and a map

$$\begin{aligned} \mathbb{Z}G \times M &\to M \\ (r, m) &\mapsto rm \end{aligned}$$

such that

1. $r(m+n) = rm + rn$,
2. $(r_1 + r_2)m = r_1 m + r_2 m$,
3. $(r_1 r_2)m = r_1(r_2 m)$,
4. $1m = m$

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A module is *trivial* if $gm = m$ for all $g \in G$, $m \in M$. We say *the trivial module* is $\mathbb{Z}$ with trivial action $gm = m$. A *free $\mathbb{Z}G$-module* on X is formal sums

$$\sum r_x x$$

for $r_x \in \mathbb{Z}G$, $x \in X$, and finitely many r_x non-zero. We will write $\mathbb{Z}G\{X\}$ for this.

Submodules and *quotient modules* are defined as we would expect.

Definition 1.3 ($\mathbb{Z}G$-map)

An *$\mathbb{Z}G$-map (or morphism)* $\alpha : M_1 \to M_2$ is a map of abelian groups, with

$$\alpha(rm_1) = r\alpha(m_1)$$

for $r \in \mathbb{Z}G$, $m_1 \in M_1$.

Example 1.4

The *augmentation map* is

$$\varepsilon : \mathbb{Z}G \to \mathbb{Z}$$
$$\sum n_g g \mapsto \sum n_g$$

where we treat $\mathbb{Z}G$ as a left $\mathbb{Z}G$-module, and $\mathbb{Z}$ as the trivial module. This is a $\mathbb{Z}G$ map, and it is also a ring homomorphism.

Notation 1.5. Write $\mathrm{Hom}_G(M, N)$ for the set of $\mathbb{Z}G$-maps from $M \to N$, with pointwise addition.

Example 1.6

If we regard $\mathbb{Z}G$ as a left $\mathbb{Z}G$-module, then

$$\mathrm{Hom}_G(\mathbb{Z}G, M) \cong M$$

for any left $\mathbb{Z}G$-module M, by sending φ to $\varphi(1)$. This is an isomorphism since

$$\varphi(r) = \varphi(r1) = r\varphi(1)$$

Note that $\mathrm{Hom}_G(\mathbb{Z}G, M)$ is a left $\mathbb{Z}G$-module, with

$$(s\phi)(r) = \phi(rs)$$

In particular,

$$\mathrm{Hom}_G(\mathbb{Z}G, \mathbb{Z}G) \cong \mathbb{Z}G$$

With this, $\varphi \mapsto \varphi(1)$ corresponds to multiplication on the *right* by $\varphi(1)$.

Note that G may not be abelian, so the distinction between left and right modules matters.

Definition 1.7

If $f : M_1 \to M_2$ is a $\mathbb{Z}G$-map, then the dual map is

$$f^* : \mathrm{Hom}_G(M_2, N) \to \mathrm{Hom}_G(M_1, N)$$

for any $\mathbb{Z}G$-module N. This is given by

$$f^*(\varphi) = \varphi \circ f$$

Similarly, if $f : N_1 \to N_2$, then we have

$$f_* : \mathrm{Hom}_G(M, N_1) \to \mathrm{Hom}(M, N_2)$$

with $f_*(\varphi) = f \circ \varphi$.

These are maps of abelian groups, since in general $\mathrm{Hom}_G(M, N)$ doesn't have to be a $\mathbb{Z}G$-module, as we don't have a right action.

Example 1.8 (Prototypical example)

Let $G = \langle t \rangle$ be an infinite cyclic group, and consider the graph Γ with vertices $\{v_i\}_{i\in\mathbb{Z}}$, edges $v_i \leftrightarrow v_{i+1}$. Then G acts on the set V of vertices, by $tv_i = v_{i+1}$, and also on the set E of edges. The action is transitive in both cases. The formal integral sums $\mathbb{Z}V$ and $\mathbb{Z}E$ are $\mathbb{Z}G$-modules, and are free. Fix the edge $e : v_0 \leftrightarrow v_1$. Then we have $\mathbb{Z}G$-maps

$$\begin{aligned} \mathrm{d} : \mathbb{Z}E &\to \mathbb{Z}V \\ e &\mapsto v_1 - v_0 \end{aligned}$$

and also

$$\begin{aligned} \mathbb{Z}V &\to \mathbb{Z} \\ v_0 &\mapsto 1 \end{aligned}$$

This correspond to the augmentation map.

Definition 1.9 (chain complex, exact, homology)

A *chain complex* of $\mathbb{Z}G$-modules is a sequence

$$M_s \xrightarrow{\mathrm{d}_s} M_{s-1} \xrightarrow{\mathrm{d}_{s-1}} \cdots \xrightarrow{\mathrm{d}_{t+1}} M_t$$

of $\mathbb{Z}G$-modules and maps, such that for $t < n < s$, $\mathrm{d}_n \circ \mathrm{d}_{n+1} = 0$. We will write

$$M_\bullet = (M_n, \mathrm{d}_n)_{t\le n\le s}$$

We say that $M_\bullet$ is *exact* at M_n if $\mathrm{im}(\mathrm{d}_{n+1}) = \ker(\mathrm{d}n)$. The sequence is exact if if it exact at all M_n with $t < n < s$.

The *homology* of $M_\bullet$ is given by

$$\begin{aligned} \mathrm{H}_s(M_\bullet) &= \ker(\mathrm{d}s) \\ \mathrm{H}_n(M_\bullet) &= \frac{\ker(\mathrm{d}_n)}{\mathrm{im}(\mathrm{d}_{n+1})} \text{ for } t < n < s \\ \mathrm{H}_t(M_\bullet) &= \mathrm{coker}(\mathrm{d}_{t+1}) \end{aligned}$$

A *short exact sequence* is a chain complex of the form

$$0 \longrightarrow M_1 \xrightarrow{\alpha} M_2 \xrightarrow{\beta} M_3 \longrightarrow 0$$

In our prototypical example, we have a short exact sequence

$$0 \longrightarrow \mathbb{Z}E \longrightarrow \mathbb{Z}V \longrightarrow \mathbb{Z} \longrightarrow 0$$

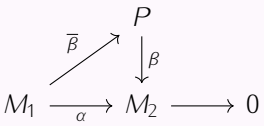
which corresponds to

$$0 \longrightarrow \mathbb{Z}G \longrightarrow \mathbb{Z}G \longrightarrow \mathbb{Z} \longrightarrow 0$$

where the map $\mathbb{Z}G \to \mathbb{Z}G$ is multiplication on the right by $t - 1$.

Definition 1.10 (projective)

A $\mathbb{Z}G$-module P is *projective* if for every surjective $\mathbb{Z}G$-map $\alpha : M_1 \to M_2$, and every $\mathbb{Z}G$-map $\beta : P \to M_2$, there exists $\overline{\beta}$ such that



commutes.

If we have any short exact sequence

$$0 \longrightarrow N \xrightarrow{f} M_1 \xrightarrow{\alpha} M_2 \longrightarrow 0$$

Consider

$$0 \longrightarrow \mathrm{Hom}_G(P, N) \xrightarrow{f_*} \mathrm{Hom}_G(P, M_1) \xrightarrow{\alpha_*} \mathrm{Hom}_G(P, M_2) \longrightarrow 0 \qquad (*)$$

Then P is projective if and only if $(*)$ is exact. Note that in $(*)$, we always have exactness except at $\mathrm{Hom}_G(P, M_2)$.

Index

$\mathbb{Z}G$-map, 2

augmentation map, 2

chain complex, 3

exact, 3

free $\mathbb{Z}G$ module, 2

$\mathrm{Hom}_G(M, N)$, 2
homology, 3

integral group ring, 1

projective, 4

short exact sequence, 3

trivial $\mathbb{Z}G$ module, 2

$\mathbb{Z}G$ module, 1